Multimedia Signal Processing - 2nd Module

July 22, 2024

1. **Question (7 pts)** Consider a discrete source X emitting the following series of symbols

$$x = \{3,\ 15,\ 9,\ 18,\ 15,\ 15,\ -3,\ 12\}.$$

You are given a quantizer $Q(\cdot)$ that applies the following quantization rule

$$Q(x) = \left\lfloor \frac{x}{\Delta} \right\rceil \cdot \Delta,$$

where $\lfloor \cdot \rceil$ rounds to the nearest integer, and $\Delta = 2$ is the quantization step.

   (a) Compute the sequence $\hat{x}_{\text{PCM}}$ obtained by encoding and decoding x using a PCM scheme.

   (b) Compute the sequence $\hat{x}_{\text{DPCM}}$ obtained by encoding and decoding x using a closed-loop DPCM scheme. The predictor simply returns the previous sample, and the first sample is encoded using PCM.

   (c) Compare the mean squared error (MSE) obtained with the PCM and with the DPCM.

   (d) Compute the Huffman code to encode $\hat{x}_{\text{PCM}}$. Estimate the needed probability mass function (p.m.f.) directly from the observations you have.

   (e) Should you compute the Huffman code to store on file the DPCM version of x, which is the alphabet that you would consider? Why?

2. **Question (4 pts)** Consider the problem of image coding.

   (a) With the help of a block diagram, describe the JPEG image encoder.

   (b) Report the steps performed by the decoder.

3. **Question (4 pts)** Considering the process of perceptual audio coding.

   (a) Explain the concept of absolute threshold of hearing.

   (b) Explain the concept of frequency masking.

   (c) Briefly report how these two concepts can be used to achieve perceptual audio coding in relation to bit allocation.

4. **Questions (8 pts - each answer can be either TRUE or FALSE - Incorrect answers give you -1 points, unanswered 0 points, correct answers 1 point - Provide a brief comment when in doubt)**

   - [T] [F] Consider two independent discrete sources X and Y with entropy $H(X) = 1.5$ and $H(Y) = 1.5$. The joint entropy must be $H(X, Y) = 1.5$.
   - [T] [F] Consider a speech coding scheme based on the use of a DPCM. Both a long-term and a short-term predictor can be used.
   - [T] [F] Consider a transform coding scheme that maps each input vector $\mathbf{x} = [x_1, x_2]$ into a vector of coefficients $\mathbf{y} = [y_1, y_2]$. The sum of the variances of x_1 and x_2 is equal to the sum of the variances of the transformed coefficients y_1 and y_2.
   - [T] [F] In a video coding scheme, motion vectors can represent motion in terms of fraction of pixels.
   - [T] [F] Considering a transform coding scheme, KLT transform is always preferable to DCT.
   - [T] [F] A scalar uniform quantizer and a Lloyd-Max quantizer provide the same solution for a uniformly distributed source.
   - [T] [F] Transform coding is always a better choice than normal PCM coding in theory and practice.
   - [T] [F] It is not possible to outperform Lloyd-Max quantization considering scalar quantizers.

5. **Question (Matlab - 10pts)**

   Consider you have an image file called `peppers.png` in your Matlab working directory. This is a RGB image with resolution 512×384.

   Data loading

   (a) Load the image and display it.

   (b) Crop the top-left 256×256 pixels of the image (use the cropped version of the image hereinafter).

   (c) Convert to grayscale and display the image again. Feel free to chose the RGB-to-gray conversion you want.

   DPCM coding

   Considering the color component you extracted from the image, implement a close-loop DPCM-based coding scheme. Consider the image as a monodimensional vector, use the previous pixel as predictor, and assume that the first pixel is known by the decoder.

   (d) Turn the two dimensional image into a monodimensional vector.

   (e) Consider that you want a quantizer with 8 possible levels, define a quantizer that is reasonably adapted to the residual dynamic (hint: use an open-loop estimate of the residual to tune your quantizer).

   (f) Encode and decode the vector using the defined DPCM and quantizer.

   (g) Turn the decoded vector back into a matrix and display the decoded image.

   (h) Compute the introduced MSE.